

VPA IB DP CS — Topic 2: Computer Organization (Weeks 10 – 18)

Topic 2: Computer Organization 计算机组成 Jìsuànjī Zǔchéng

IB Syllabus: Topic 2 (2.1 – 2.5) | Level: SL + HL | Duration: 9 weeks (30 hours)

Topic 2.1: Computer Architecture 计算机体系结构 Jìsuànjī Tǐxì Jiégòu

Week 10 — CPU Components CPU组件 CPU Zǔjiàn

Lesson	Content	Activities	Differentiation
1	The CPU: ALU, CU, registers (PC, IR, MAR, MDR, ACC) (算术逻辑单元, 控制单元, 寄存器)	Label CPU diagram; match register to function	Bronze: Label diagram; Silver: Explain function; Gold: Trace data flow during execution
2	The Fetch-Decode-Execute (FDE) cycle (取指-译码-执行周期)	Step-through animation; manual trace on worksheet	Bronze: Identify stages; Silver: Describe each stage; Gold: Explain interrupt handling
3	FDE cycle continued: role of each register at each stage	Trace table: show register contents cycle by cycle	Scaffolded trace tables with varying detail
4	Bus systems: data bus, address bus, control bus (总线 zǒngxiàn)	Calculate address bus width from memory capacity	Bronze: Identify buses; Silver: Calculate width; Gold: Discuss bottleneck implications
5	TOK: Is the CPU a model of human thought, or is it a fundamentally different kind of processor?	Discussion + written reflection	All: Contribute; Gold: Reference cognitive science

TOK Knowledge Question: The Fetch-Decode-Execute cycle is a mechanistic description of how computers "think." To what extent does this mechanical process illuminate—or obscure—what it means to "process information"? Can a machine truly "understand" what it processes?

Week 11 — Memory and the Stored Program Concept

Lesson	Content	Activities
1	Von Neumann architecture: stored program concept, sequential execution (冯·诺依曼架构, 存储程序概念)	Compare Von Neumann to Harvard architecture
2	Machine code vs assembly language vs high-level languages	Translate a simple program through all three levels
3	Interrupts: purpose, interrupt service routine, vectored interrupts	Scenario: handle multiple interrupt sources
4	Clock speed, word length, cores, cache size: impact on performance	Compare CPUs using specification sheets; justify purchase

Lesson	Content	Activities
5	Formative: CPU architecture quiz (20 marks, 25 minutes)	Timed quiz; instant feedback

Topic 2.2: Memory and Storage 内存与存储 Nèicún yǔ Cúncǔhǔ

Week 12 — Primary Memory 主存储器 Zhǔ Cúncǔhǔ Qì

Lesson	Content	Activities
1	RAM: DRAM, SRAM, volatile memory (随机存取存储器)	Compare RAM types; explain why RAM is volatile
2	ROM: Mask ROM, PROM, EPROM, EEPROM, BIOS (只读存储器)	Research: what is stored in your computer's BIOS?
3	Cache memory: L1, L2, L3; spatial and temporal locality	Analogy exercise: cache as desk, RAM as bookshelf, HDD as library
4	Virtual memory: paging, page fault, thrashing (虚拟内存, 分页)	Simulate paging with physical counters; observe thrashing
5	Consolidation: Memory hierarchy poster task	Create visual hierarchy with speed, capacity, cost per bit

Week 13 — Secondary Storage 辅助存储器 Fǔzhù Cúncǔhǔ Qì

Lesson	Content	Activities
1	Magnetic storage: HDD structure, platters, read/write heads, seek time, latency (硬盘 yìngpán)	Calculate average access time; discuss fragmentation
2	Optical storage: CD, DVD, Blu-ray; reflective layers, pit/land	Compare capacities and use cases
3	Solid-state storage: flash memory, NAND, SSD advantages/disadvantages	Debate: SSD vs HDD for different applications
4	Emerging storage: MRAM, optical holographic, DNA storage (HL discussion)	Research presentation on future storage
5	Formative: Memory and storage comparison table (graded for accuracy and completeness)	Individual submission

Topic 2.3: Operating Systems and Application Software 操作系统与应用软件 Cāozuò Xìtǒng yǔ Yìngyòng Ruǎnjiàn

Week 14 — Operating Systems 操作系统 Cāozuò Xìtǒng

Lesson	Content	Activities
1	OS functions: user interface, memory management, peripheral management, multitasking, security (操作系统功能)	Map OS functions to user experiences

Lesson	Content	Activities
2	Types of OS: batch, real-time, distributed, embedded, multi-user	Classify OS examples; justify classification
3	Application software: general-purpose, specialist, custom-written, utility software	Audit software on a lab computer; categorise each
4	Software licensing: proprietary, open source, freeware, shareware, EULA (软件许可 ruǎnjiàn xǔkě)	Ethical discussion: piracy, FOSS, digital rights
5	TOK: Is software "real" in the same way hardware is? What kind of existence does code have?	Philosophy of software discussion

TOK Knowledge Question: Software controls hardware, yet software itself is intangible—a pattern of 1s and 0s. In what sense can something non-physical cause physical effects? Does this challenge our standard categories of existence?

Topic 2.4: Binary Representation 二进制表示 □rjìnzhì Biǎoshì

Week 15 — Number Systems 数制系统 Shùzhì Xìtǒng

Lesson	Content	Activities	Differentiation
1	Binary, denary, hexadecimal: conversion between all three (二进制, 十进制, 十六进制)	Conversion drills; bingo	Bronze: Binary□Denary; Silver: +Hex; Gold: Fractional binary
2	Binary addition and overflow; two's complement representation (补码 bǔmǎ)	Binary addition race; detect overflow	Bronze: Positive numbers; Silver: Two's complement; Gold: Subtraction via addition
3	Binary shifts: logical and arithmetic; multiplication/division by powers of 2	Worksheet: predict result of shifts	Bronze: Left shift; Silver: Right shift; Gold: Arithmetic shift with sign preservation
4	Floating-point representation: mantissa and exponent, normalisation (浮点数 fúdiǎn shù)	Convert denary to normalised floating-point	Bronze: Understand concept; Silver: 8-bit conversions; Gold: Discuss precision trade-offs
5	Truncation errors, rounding errors, underflow, overflow (截断错误 jiéduàn cuòwù)	Calculate errors in given examples; discuss real-world consequences	All: Calculate; Gold: Research Patriot missile / Ariane 5 case studies

Week 16 — Data Representation 数据表示 Shùjù Biǎoshì

Lesson	Content	Activities
1	Character encoding: ASCII, Unicode (UTF-8, UTF-16) (字符编码 zìfú biānmǎ)	Encode/decode messages; explore Unicode for Chinese characters
2	Colour representation: RGB, bit depth, resolution; bitmap vs vector	Calculate file size of images; compare formats
3	Sound representation: sampling rate, bit depth, Nyquist theorem; MIDI	Calculate audio file sizes; discuss compression

Lesson	Content	Activities
4	Data compression: lossy vs lossless; RLE, Huffman coding (HL)	Compress text using RLE; compare before/after
5	Formative: Binary calculations test (30 marks, 40 minutes)	Timed test; peer marking

Topic 2.5: Simple Logic Gates 基本逻辑门 Jīběn Luójí Mén

Week 17 — Boolean Logic 布尔逻辑 Bù'ěr Luójí

Lesson	Content	Activities	Differentiation
1	Logic gates: AND, OR, NOT; symbols, truth tables (与门, 或门, 非门)	Build truth tables; match symbols to names	Bronze: Complete given tables; Silver: Construct from description; Gold: Simplify expressions
2	NAND, NOR, XOR, XNOR (与非门, 或非门, 异或门)	Complete all 7 truth tables; identify from timing diagrams	Bronze: 3 basic gates; Silver: All 7; Gold: Universal gate proofs
3	Constructing logic circuits from Boolean expressions	Draw circuit diagrams; use logic simulator software	Bronze: 2-gate circuits; Silver: 3-4 gates; Gold: Multi-level optimisation
4	Deriving Boolean expressions from truth tables (sum of products)	Given a scenario, create truth table, expression, circuit	Scaffolded worksheets
5	Simple logic applications: half-adder, full-adder (HL), security system design	Design a 2-floor burglar alarm system	All: Participate; Gold: Design full-adder and explain carry propagation

Week 18 — Topic 2 Review and Semester 1 Examination

Lesson	Content	Activities
1	Revision: Computer architecture and memory hierarchy	Flashcard relay; concept connections
2	Revision: Binary representation and logic gates	Calculation practice; circuit design challenge
3	Semester 1 Examination (Paper 1 mock, SL 1h30 / HL 2h10)	Formal examination conditions
4	Examination feedback and DIRT (Dedicated Improvement Reflection Time)	Markscheme analysis; target setting
5	Semester 1 roundup; IA client proposal check	Submit IA client proposal draft; celebrate progress

Key Vocabulary (Topic 2)

English	中文	Pinyin	Definition
CPU	中央处理器	zhōngyāng chǔlǐ qì	The main processing unit of a computer; executes instructions
ALU	算术逻辑单元	suànshù luójí dānyuán	Performs arithmetic and logical operations
Register	寄存器	jìcúqì	High-speed storage locations within the CPU
Bus	总线	zǒngxiàn	A communication pathway for data between components
RAM	随机存取存储器	suíjī cúnqǔ cúnchǔ qì	Volatile primary memory for currently running programs
ROM	只读存储器	zhǐdú cúnchǔ qì	Non-volatile memory containing permanent instructions
Cache	高速缓存	gāosù huǎncún	Fast memory between CPU and RAM for frequently used data
Virtual memory	虚拟内存	xūnǐ nèicún	Using disk space to extend apparent RAM capacity
Two's complement	补码	bǔmǎ	A method for representing signed integers in binary
Logic gate	逻辑门	luójí mén	An electronic circuit performing a Boolean function
Mantissa	尾数	wěishù	The significant digits in a floating-point number
Exponent	指数	zhǐshù	The power to which the base is raised in floating-point
Interrupt	中断	zhōngduàn	A signal suspending current processing for urgent handling
Word length	字长	zìcháng	The number of bits processed by the CPU in one operation

Teacher Checklist — End of Topic 2:

- ☐ All IB 2.1–2.5 syllabus points covered
- ☐ Binary calculations practised to fluency
- ☐ Logic gate truth tables memorised
- ☐ TOK connections documented
- ☐ IA client proposals submitted and approved
- ☐ Semester 1 examination sat and reviewed
- ☐ Victoria Park Academy Shenzhen | IB DP Computer Science | Topic 2 Scheme of Work